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Sero-prevalence and associated factors of transfusion-transmitted infections of HBV, HCV, HIV, and syphilis among voluntary blood donors in Gondar District Blood Bank, Ethiopia: a retrospective study

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Abstract

Background Blood transfusion is a life-saving intervention performed worldwide; however, it remains a significant public health concern due to the risk of transfusion-transmitted infections (TTIs). Globally, millions are infected with hepatitis B virus (HBV), hepatitis C virus (HCV), human immunodeficiency virus (HIV), and *Treponema pallidum* (syphilis) through unsafe blood transfusion practices. In resource-limited settings like Ethiopia, ensuring blood safety is especially challenging due to inadequate screening and surveillance systems. This study aimed to assess the prevalence and associated risk factors of HBV, HCV, HIV, and syphilis among voluntary blood donors in the North Gondar District Blood Bank, Northwest Ethiopia.

Method A retrospective cross-sectional study was conducted using data from 265 voluntary blood donors at the North Gondar District Blood Bank, collected between February 15, 2021, and May 24, 2022. Data were retrieved from the blood bank records between June 1 and June 15, 2022. Donor sociodemographic characteristics and transfusion-transmitted infection (TTI) screening results were extracted from these records. Descriptive statistics were used to estimate prevalence. Variables with a p-value < 0.25 in Bivariable analysis were included in a multivariable logistic regression model to identify independent predictors of TTIs. Multicollinearity was assessed using the variance inflation factor (VIF), and model fitness was evaluated using the Hosmer-Lemeshow goodness-of-fit test. A p-value < 0.05 was considered statistically significant.

Result The study covered 265 blood donors in the Gondar blood bank district between February 15, 2021, and May 24, 2022. The total sero-prevalence of HIV, HCV, HBV and syphilis during the follow-up period was 2.26%, 2.64%, 4.53%, and 7.17%, respectively. Age groups of 26–35 for blood donors screening for HCV (AOR = 2.1, 95% CI 1.1, 3.29), Male donors (AOR = 1.2, 95% CI 1.10, 6.52), age groups of 26–35 for screening for HBV (AOR = 1.1, 95% CI 1.00, 1.24), age groups greater than 45 for screening for Treponema pallidum (AOR = 8.6, 95% CI 7.44, 10.02), and divorced marital

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status (AOR = 2.1, 95% CI 1.64, 6.94) were significantly associated with the prevalence of Sero-prevalence trans-fusion transmissible infections occurrence.

Conclusion This study found that the overall sero-prevalence of TTIs among blood donors in North Gondar was notable, with HCV (4.53%) and syphilis (7.17%) being the most prevalent. Statistically significant associations were observed between HCV and the 26–35 age group, syphilis and donors aged above 45, HBV and the 26–35 age group, male gender with TTIs, and divorced marital status with syphilis. These findings underscore the need for more targeted donor screening strategies focusing on these high-risk groups to enhance transfusion safety. Further studies with larger sample sizes are recommended to validate these associations and explore additional risk factors.

Clinical trial number was not applicable in the study.

Keywords Sero-prevalence, HBV, HCV, HIV, Syphilis, Blood donors, Ethiopia

Introduction

Transfusion-transmitted infections (TTIs) remain a global public health concern, particularly in low- and middle-income countries where blood screening systems are often under-resourced [1]. The burden of TTIs varies by region, with the World Health Organization estimating that the prevalence of HBV among blood donors is around 6% in Africa, compared to less than 1% in high-income countries [1, 2]. Similarly, the prevalence of HCV and HIV among donors in sub-Saharan Africa remains significantly higher than in other parts of the world due to endemic infection rates and gaps in health infrastructure [3, 4]. In Ethiopia, previous studies have reported TTI prevalence rates ranging from 3% to over 10% among blood donors, with variations by geographic region, donor type, and demographic characteristics [5].

Despite national guidelines requiring mandatory screening of donated blood for HIV, HBV, HCV, and syphilis, the effectiveness of screening practices at the ground level is often compromised by multiple factors [6]. These include inadequate supply of test kits, use of rapid diagnostic tests with lower sensitivity, poor quality assurance mechanisms, insufficient training of laboratory staff, and lack of confirmatory testing protocol [7, 8].

Blood transfusion services (BTS) are a vital component of modern healthcare, providing life-saving support in the management of medical, surgical, and obstetric emergencies [1]. They play a critical role in reducing morbidity and mortality by replacing lost blood components and correcting hematologic deficiencies [9]. Globally, it is estimated that pregnancy-related hemorrhage accounts for approximately 25% of maternal deaths, while anemia contributes to up to 15% of childhood mortality in sub-Saharan Africa [10, 11]. In Ethiopia, high rates of maternal mortality, trauma from road traffic accidents, and complications from infectious diseases continue to drive the demand for safe and accessible blood transfusion services [12]. Transfusions are essential in the management of conditions such as severe anemia, thrombocytopenia, hypoalbuminemia, and coagulation disorders, particularly in settings with limited advanced medical interventions [6].

Numerous studies have been conducted globally to assess the prevalence and patterns of TTIs among blood donors, identifying both rising and declining trends [13]. These findings emphasize the need for robust screening systems within blood transfusion units, which also serve as indirect indicators of the prevalence of these infections in otherwise healthy populations [8, 13].The responsibility for ensuring safe blood transfusion falls on healthcare institutions, which must rigorously screen donors and test collected blood [14, 15]. While laboratory-based screening and careful donor selection have significantly improved safety, challenges persist [3]. These include limitations of screening assays during the diagnostic window period, the emergence of immunologic variants, non-seroconverting carriers, inadequate quality assurance in testing, and resource constraints such as funding and trained personnel [16–18].

While previous studies have assessed TTIs among Ethiopian blood donors, many are either outdated or focused on limited regions, lacking detailed analysis of associated risk factors such as age, sex, marital status, or behavioral characteristics [3, 19]. Moreover, screening practices and their limitations at the district level have not been well documented. Understanding the current seroprevalence of TTIs and identifying demographic risk factors is essential for developing targeted interventions to improve transfusion safety.

Therefore, this study was conducted to assess the seroprevalence of HBV, HCV, HIV, and syphilis and identify associated risk factors among voluntary blood donors at the North Gondar District Blood Bank, Northwest Ethiopia. The evidence generated is expected to contribute to improving blood safety protocols and guide region-specific preventive strategies.

Materials and methods

Study design, period and setting

The North Gondar District Blood Bank conducted a retrospective cross-sectional analysis of blood donors who

donated between February 15, 2021, and May 24, 2022, with a total of 1,561 donors during this period. The blood bank is located in Gondar, a town situated 738 km northwest of Ethiopia's capital, Addis Ababa. It serves a catchment population of approximately five million people in North Gondar and the surrounding areas. On average, the blood bank collects about 2,500 units of blood annually. Most of these blood units are used for gynecological, surgical, and emergency medical cases.

Population

Source and study population All individuals aged 18–65 years who donated blood at the Gondar Blood Bank between February 15, 2021, and May 24, 2022.

Eligibility

Inclusion criteria Individuals aged 18 to 65 years who donated blood at the Gondar Blood Bank between February 15, 2021, and May 24, 2022, with complete and documented data on demographic and serological test results.

Exclusion criteria Individuals were excluded if they donated blood outside the study period, had incomplete or missing data, were repeat donors (only the first donation was included), or had documented deferrals due to health or risk factors at the time of donation.

Pretest and tool validation

The data collection tools (structured questionnaires and checklists) were pretested on 14 individuals (5% of sample size) at Azezo Health Center in Gondar. The pretest assessed clarity, flow, and reliability. The internal consistency of the tool was evaluated using Cronbach's alpha, which yielded a value of 0.82, indicating good reliability. Necessary modifications were made based on pretest findings before the main data collection.

Study variables

Dependent variable Prevalence Hepatitis B, C, HIV virus and syphilis.

Independent Variable: Sociodemographic variables: age, sex, educational status, occupational status, marital status, residence, blood transfusion,

Operational definition

Blood donor An individual aged 18–65 years who voluntarily donates blood after meeting national eligibility screening criteria [20].

Volunteer blood donor A person who donates blood freely without receiving any form of compensation [20].

Blood Bank: A bank that give blood transfusion service Volunteer Blood Donor: People who give blood without any payment [21].

Unsafe therapeutic injection An injection administered by a non-health professional using unsterile equipment or improper technique [22].

Transfusion-Transmitted infections (TTIs) Infections including HBV, HCV, HIV, and syphilis that are transmissible through blood transfusion, confirmed using standard diagnostic techniques [21].

Diagnostic approaches

Hepatitis B Virus (HBV): Detection of hepatitis B surface antigen (HBsAg) using Enzyme-Linked Immunosorbent Assay (ELISA). A positive HBsAg test indicates active HBV infection **Hepatitis C Virus (HCV)**: Detection of anti-HCV antibodies using ELISA. A reactive test suggests current or past HCV infection and requires further confirmatory testing.

HCV Detected using ELISA for anti-HCV antibodies.

Human immunodeficiency virus (HIV) Screening follows the national testing algorithm, starting with an initial rapid immunochromatographic assay. Reactive samples are confirmed by a second rapid test and, if necessary, a third confirmatory test.

Syphilis Diagnosed using the Rapid Plasma Reagin (RPR) test. Reactive results indicate probable active infection and may be confirmed by treponemal tests where available.

Sample size calculation

Sample size was calculated using the single population proportion formula with 95% confidence interval, 3% margin of error, and an assumed prevalence of 6.0% from a previous study [17, 23].

$$n = \frac{\left(\frac{z}{2}\right) 2\,(\mathrm{p})\,(1-\mathrm{p})}{\mathrm{d2}}$$

$$= \; n = 1.96^2 * 0.06 \frac{(1-0.06)}{(0.03)^2} = 241$$

P = prevalence of HBV, HCV. HIV and syphilis among blood donors in Bahir Dar blood bank district [39]. Adding a 10% non-response rate: nf = 241+ (241 × 0.10) = 265. Thus, the final sample size was 265.

Sampling technique

From a total of 1,561 donors during the study period, systematic random sampling was employed with a sampling interval of 6. The first donor was randomly selected between 1 and 6, and every 6th donor thereafter was included until the sample size was reached (Fig. 1).

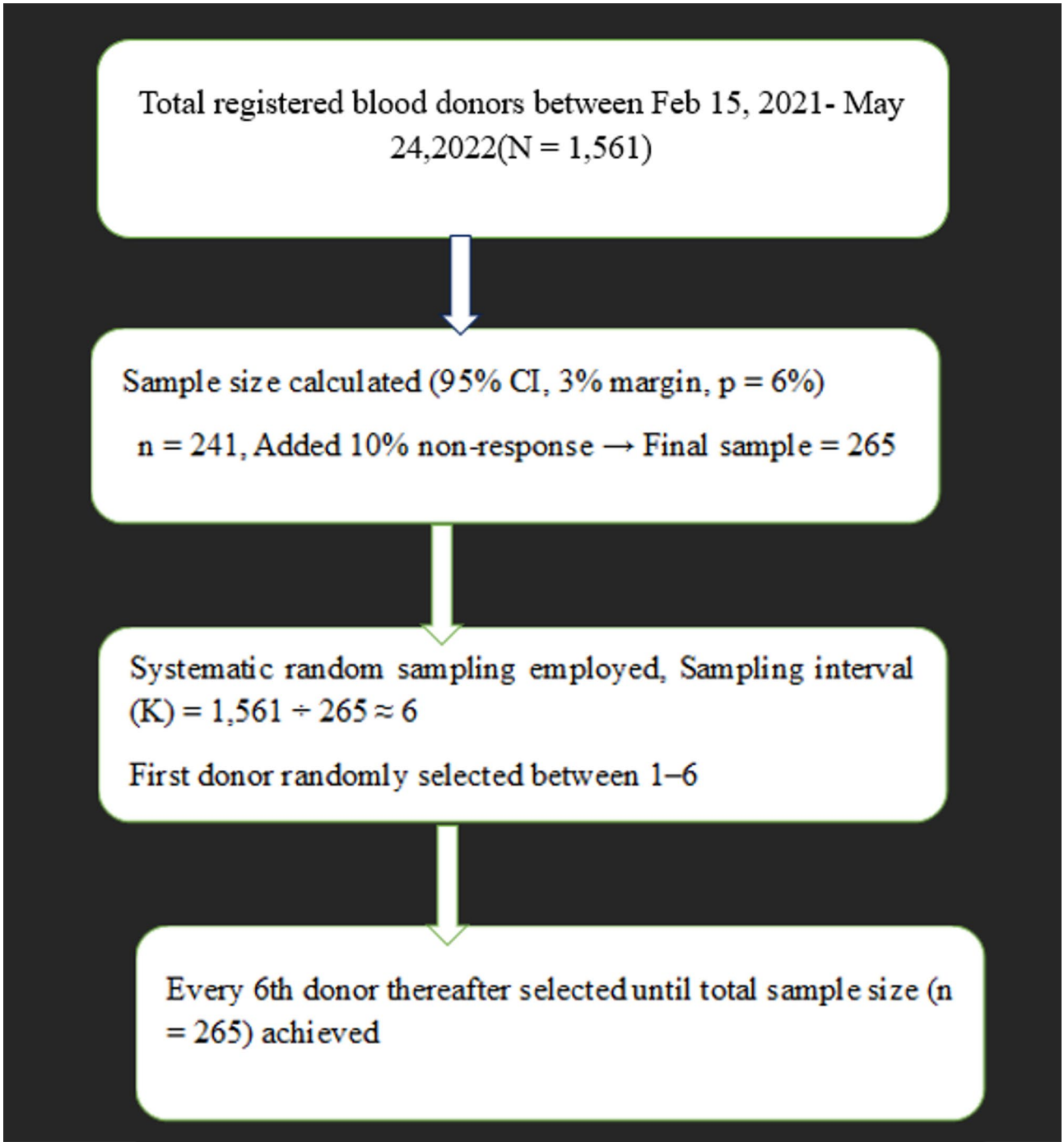


Fig. 1 Schematic diagram for the sampling procedure among blood Donors in North Gondar District, Ethiopia (2021–2022)

Data collection

The research data were collected using structured questionnaires and checklists, which were designed based on a review of similar studies published previously [8, 24–27]. Data on sociodemographic characteristics and serological results were extracted retrospectively from blood bank records using structured checklists. A total of 1561 volunteer blood donors donated blood throughout the study period. After calculating K, or K = 1561/265 = 5.9 = ~ 6, we want to transfer data from the register to those who fulfill the inclusion criteria. In this case, k is the number to which we want to take data. The total number of blood donors in a given time period is 1561, and our sample size is 265.

Laboratory test and result

Blood samples were screened for transfusion-transmitted infections using standardized assays. **Hepatitis B virus (HBV)**: was detected through hepatitis B surface antigen (HBsAg) testing using the Murex HBsAg ELISA kit (Murex Biotech Ltd. UK), with reported sensitivity and specificity greater than 99% [28].

Hepatitis C virus (HCV) screening involved detection of anti-HCV antibodies using the Abbott ARCHITECT Anti-HCV ELISA kit (Abbott Laboratories, USA), which has sensitivity and specificity above 98% [29].

Human immunodeficiency virus (HIV) screening followed the Ethiopian national testing algorithm, starting with the SD Bio line HIV-1/2 3.0 rapid test (Standard Diagnostics Inc., Korea) and confirmation of reactive results using the Unit-Gold HIV test (Trinity Biotech, Ireland); both tests have sensitivity and specificity exceeding 99% [30].

Syphilis screening was conducted using the Rapid Plasma Reagin (RPR) test (BD Diagnostics, USA), and reactive samples were confirmed using treponemal-specific tests where available [31].

Quality assurance

To ensure the accuracy and reliability of laboratory results, all laboratory personnel were trained and strictly adhered to standard operating procedures recommended by both the manufacturers and the World Health Organization (WHO). Positive and negative controls were included with each batch of tests to validate assay performance. All initially reactive samples were retested to confirm the results. Regular maintenance and calibration of laboratory equipment were conducted according to established guidelines. Additionally, data extraction was carefully double-checked to ensure completeness and accuracy.

Data management and analysis

Data were initially entered and cleaned using Epi Info version 7 and then exported to SPSS version 25 for further analysis. Descriptive statistics were performed to summarize the sociodemographic characteristics and the prevalence of transfusion-transmissible infections (TTIs). Bivariate logistic regression was conducted to identify candidate variables with a p-value less than 0.25 for inclusion in the multivariable logistic regression model. The multivariable logistic regression was then used to determine factors independently associated with TTIs. To assess the adequacy of the logistic regression model, the Hosmer-Lemeshow goodness-of-fit test was applied, where a p-value greater than 0.05 indicated an acceptable fit and Variables that did not meet the required fit were excluded from the multivariate logistic analysis model [32]. Multicollinearity was assessed using the variance inflation factor (VIF), and model fitness was evaluated using the Hosmer-Lemeshow goodness-of-fit test. Statistical significance was declared for an adjusted odds ratio (AOR) with a 95% confidence interval at $P < 0.05$.

Results

Socio demographic characteristics

The study analyzed records from 265 blood donors at the North Gondar District Blood Bank between February 2021 and May 2022. Among the donors, 177 (66.8%) were male. The age of participants ranged from 18 to 54 years, with a median age of 33 years (SD = 8.7). The largest age group was 26–35 years, comprising 107 donors (40.4%), while those over 45 years constituted the smallest group with 21 donors (7.9%). Regarding occupation, students accounted for the largest proportion at 71 donors (26.8%), followed by government employees with 58 donors (21.9%). Most donors (206, 77.7%) were from urban areas, with students representing nearly two-thirds of this group. Educational attainment varied, with 83 donors (31.3%) holding a TVET diploma and 58 donors (21.9%) possessing a degree or higher (see Table 1).

Sero-prevalence of transfusion transmissible infections

HIV prevalence was highest among farmers (33.3%) and students (16.7%). The 26–35 age group exhibited the greatest prevalence for HIV (41.7%), HBV (50%), and syphilis (47.4%). Male donors had higher infection rates compared to females. HIV was more prevalent in rural donors, whereas HBV, HCV, and syphilis were more common in urban donors).

The sero-prevalence rates of TTIs among blood donors were: HBV 4.53% (12/265; 95% CI: 2.37%–7.74%), HCV 2.64% (7/265; 95% CI: 1.07%–5.36%), HIV 2.26% (6/265; 95% CI: 0.83%–4.87%), and syphilis 7.17% (19/265; 95% CI: 4.39%–10.95%) (Table 2).

Prevalence of transfusion-transmissible infections among blood donors

The study conducted at the North Gondar District Blood Bank revealed notable sero-prevalence rates of transfusion-transmissible infections among voluntary blood donors. The overall prevalence rates were 2.26% for HIV, 4.53% for HBV, 2.64% for HCV, and 7.17% for syphilis. Syphilis showed the highest prevalence among the infections studied. Significant associations were observed with certain demographic factors, including higher infection rates in male donors and those aged 26–35 for HCV and HBV, as well as increased syphilis prevalence in donors

Table 1 Sociodemographic characteristics of blood donors in North Gondar district blood Bank, Ethiopia ($n=265$)

Variables	Category	Frequency	Percent
Sex	Male	177	66.79
	Female	88	33.21
Age	18–25	62	23.40
	26–35	107	40.38
	36–45	75	28.30
	>45	21	7.92
Residence	Rural	59	22.26
	Urban	206	77.74
Education status	Non formal education	29	10.94
	1–8	42	15.85
	9–12	53	20.00
	TVET diploma	83	31.32
	Degree and above	58	21.89
Occupation	Student	71	26.79
	Government employee	58	21.89
	Private employee	41	15.47
	Daily laborer	36	13.58
	Farmer	38	14.34
	Other	21	7.92
Religion	Orthodox	128	48.30
	Muslim	52	19.62
	Protestant	29	10.94
	Other	56	21.13
Blood type	A	83	31.32
	B	69	26.04
	AB	23	8.68
	O	90	33.96
Marital status	Single	92	34.72
	Married	138	52.08
	Divorced	30	11.32
	Windowed	5	1.89

Table 2 Sero-prevalence of HIV, HBV, HCV and syphilis among blood donors in North Gondar district blood Bank, Ethiopia from February 15, 2021 to May 24, 2022 ($n=265$)

Variable	Frequency	Percent (%)	95% Confidence Interval
HBV	12	4.53	3.37–7.74
HCV	7	2.64	1.07–5.36
HIV	6	2.26	0.83–4.87
Syphilis	19	7.17	4.39–10.95

over 45 years old. These findings underscore the need for targeted screening and prevention strategies.

Focusing on high-risk to enhance blood safety in the region (Fig. 2).

Factors associated with TTIs

This study found that being in the 26–35 age group significantly increased the likelihood of having HIV, HBV, HCV, or Treponemal pallidum infections. The probability of detecting HIV and syphilis in blood donors also increased with age. Donors in the 26–35, 36–45, and

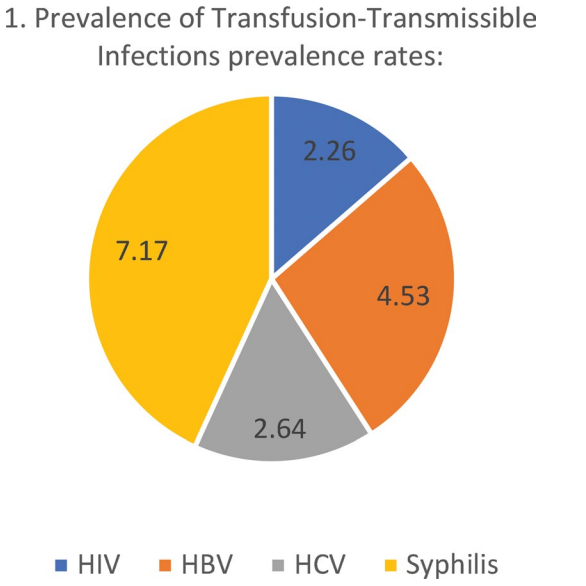


Fig. 2 Prevalence of transfusion-transmissible infections by age group and sex among blood donors in north Gondar District, Ethiopia (2021–2022)

over 45 age groups had 1.2 (1.17, 1.60), 1.6 (1.57, 8.50), and 2.7 (1.18, 3.07) times higher odds, respectively, of testing positive for HIV compared to those in the 18–24 age range. Similarly, the prevalence of Treponema pallidum (syphilis) also increased with age, with significant differences in the odds within each 10-year age group when compared to the baseline age group [18–24]. Furthermore, the prevalence of HCV was associated with age, with donors aged over 45 and those aged 26–35 having odds of 1.6 (1.5, 2.75) and 2.1 (1.1, 3.29), respectively (Table 3).

Discussion

This study found that the overall Sero-prevalence of transfusion-transmissible infections (TTIs) among blood donors in North Gondar was notable, with HBV at 4.53%, HCV at 2.64%, HIV at 2.26%, and syphilis at 7.17%. These findings highlight the ongoing public health concern of TTIs in the region.

The prevalence of HIV among blood donors in this study was 2.26%, which is higher than reports from Jigjiga (0.1%) and blood banks in Hawassa and Yirgalem Hospital (1.6%) [33–36]. This rate is also above the 1.4% Sero prevalence reported in eastern Ethiopia) [37, 38]. However, compared to other African countries, our observed prevalence remains relatively low, with reports ranging from 4.1% in Cameroon to 8.5% in Mozambique [39, 40]. Similar prevalence rates have been reported in Eritrea (2.58%) [25]. These differences may reflect regional variations in HIV risk behaviors, donor self-awareness of infection status, and sample size differences across studies.

Table 3 Multivariable logistic regression analysis for factors associated with transfusion transmission diseases: blood donors in North Gondar district blood Bank, Ethiopia from February 15, 2021 to May 24, 2022 ($n=265$)

Category		HCV AOR [95%CI]	HBV AOR [95%CI]	HIV AOR [95%CI]	Treponema pallidum AOR [95%CI]
Age	18–25	1.00	1.00	1.00	1.00
	26–35	2.1[1.1, 3.29] **	1.1[1.00, 1.24] **	1.2[1.17, 1.60] *	1.3[1.12, 10.21] ***
	36–45	0.62[2.5, 3.0]	1.3[0.23, 1.52]	1.6[1.57, 8.50] *	3.0[2.6, 6.29] *
	Greater than 45	1.6[1.5, 2.75] *	1.3[0.9, 1.84]	2.7[1.18,3.07)] *	8.6[7.44, 10.02] **
Marital status	Single	1.00	1.00	1.00	1.00
	Married	0.29(0.01, 18.82	0.76(0.14,4.20)	0.64(0.60,1.48)	1.79(0.13,18.4)
	divorced	1.3(0.23, 3.42)	2.4(1.54, 5.34) *	1.2(1.11, 3.21) ***	2.1(1.64, 6.94) **
	Windowed	0.19(0.03,1.16)	0.34(0.07,1.65)	1.00(0.61,1.64)	1.16(0.08,1.45)
Residence,	Rural	1.00	1.00	1.00	1.00
	Urban Urbana	0.77(0.45,1.18)	2.07(0.22,19.34)	0.57(0.37,0.87)	1.22(0.14,11.65)
Occupation	Student	1.00	1.00	1.00	1.00
	Gov*t employee	1.14(0.37,3.49)	0.97(0.86, 1.10)	1.14(0.37,3.49)	1.2[0.15, 2.90]
	Private employee	0.47(0.16,1.39)	0.91(0.80,1.04)	0.47(0.16,1.39)	1.1[0.48, 2.62]
	Daily laborer	0.99(0.36,2.71)	0.87(0.77, 0.98) *	0.99(0.36,2.71)	1.2[1.02, 1.49] *
	Farmer	0.19(0.03,1.16)	0.92(0.79,1.08)	1.1[0.81, 14.32]	1.7[1.17, 7.98]
	Other	0.24(0.19, 0.61)	1.00(0.87, 1.15)	0.9[0.61, 1.46]	1.3[0.53, 12.36]
Sex	Female	1.00	1.00	1.00	1.00
	Male	1.2[1.10, 6.52] **	1.4[0.04, 12.94]	0.9[0.82, 1.00]	8.9[2.69,20.14]

Where * was statically significant at $p<0.05$, ** statically significant at $p<0.02$, and ***statically significant at $p<0.01$

The sero prevalence of hepatitis B virus (HBV) in this study was 4.53%. Previous Ethiopian studies report a wide range of HBV prevalence among blood donors from 3.6% to 3.7% in Gondar and Dire dawa [23, 41] to as high as 10.9% in Jigjiga and 6.6% in Bahir Dar [8, 34]. Our findings are lower than HBV rates reported in several African countries, such as Nigeria, Mozambique, and Equatorial Guinea, where prevalence rates exceed 10% [8, 42]. The socio-cultural difference, variances in population risks, high endemicity for HBV in most African countries might be the possible reasons for these differences.

The seroprevalence of Hepatitis C virus (HCV) in our study was 2.64%, which is notably higher than local reports from Gondar, Jigjiga, and Hawassa, where prevalence ranged from 0.6% to 0.8% [8, 43, 44]. This discrepancy may be attributed to regional variations in HCV epidemiology, differences in population-level risk factors such as unsafe medical practices or substance use, and unequal access to screening, care, and support services Conversely, our findings are lower than those from Equatorial Guinea (3.7%) [8], Kenya (3.2%), and Nigeria (2.8%) [8, 45]. Others TTIs, the reason for the difference could be attributed to the prevalence of HCV in the general population, difference in risk factors, and access to screening, care and support facilities.

The overall sero-prevalence of syphilis among blood donors in this study was 7.17%. A study done in Addis Ababa National Blood Bank showed that prevalence of syphilis seropositive among blood donors as 1.3% [46]. The finding of this study was lower than some other studies done in Ethiopia. At Jigjiga and Hawassa blood bank, the syphilis prevalence was reported as 0.7% & 0.8% respectively [8, 47]. When this finding is compared to other African Countries, it is lower than in a study done in Equatorial Guinea that reported syphilis prevalence as 21.5% [8]. In addition, studies done in Cameroon and Kenya revealed that syphilis seropositive among blood donors were 8.1% and 1.2% respectively [38, 48, 49]. The possible reason for the discrepancy could be duration of the study, cultural and behavioral differences of the study participants of different countries.

From a theoretical perspective, the Health Belief Model may help explain differences in TTI prevalence. Donors with limited perceived susceptibility to infections or low health literacy may not see the need for voluntary testing or lifestyle change. Similarly, social determinants of health such as education, occupation, and access to healthcare likely influence infection risk.

In this study, old age was found to be highly associated with higher odds of occurrence HIV and syphilis in blood donors in Ethiopia. Particularly the likelihood of syphilis consistently increased with age. Similarly, a study conducted in Harare, also documented that old age was associated with higher probability of having syphilis compared to younger age [14]. Another study done in Bahir-Dar had also showed that old age was again associated with prevalence of HBV in blood donor [41]. Evidence from Eretria and Tanzania also support association of age and sero-positivity of TTIs [27, 50]. As mature age faces more encounters of exposure to the infections and live with the pathogens for longer time, the prevalence might accumulate at older age. Sex was also found to be associated with prevalence of TTIs, the prevalence was found

to be higher in males. A study conducted in Bahir-Dar had published similar finding [16].

Daily laborers had notably higher odds of infection with HBV and syphilis, likely reflecting limited access to healthcare and preventive education [16]. Furthermore, there was a noteworthy correlation between the divorce status and the occurrence of HIV, HBV, and syphilis. This may be related to people's psychological issues, any sexual problems they may have, the challenges they may have managing their homes, which could lead to occupational danger and potential assault.

These findings have critical implications for both local and global public health. Locally, targeted interventions are needed to improve donor screening and reduce TTI transmission. Globally, our data contribute to the broader understanding of regional disparities in blood safety and inform WHO strategies for improving transfusion safety in resource-limited settings.

The retrospective cross-sectional design limits causal inferences between risk factors and infection status. Relying on secondary data introduces potential errors, missing values, and inconsistent records. Since data were collected from a single blood bank, the findings may not be generalizable to other regions or settings in Ethiopia. Additionally, selection bias is possible as blood donors might represent a healthier subset of the population.

Conclusion

This study found that the overall sero-prevalence of TTIs among blood donors in North Gondar was notable, with HCV (4.53%) and syphilis (7.17%) being the most prevalent. Statistically significant associations were observed between HCV and the 26–35 age group, syphilis and donors aged above 45, HBV and the 26–35 age group, male gender with TTIs, and divorced marital status with syphilis. These findings emphasize the need for targeted screening strategies, strengthened donor selection protocols, and community education focused on high-risk groups to improve blood safety and reduce the spread of TTIs.

Recommendation: Based on the findings, it is recommended that policymakers prioritize risk-based screening, expand diagnostic capacity for TTIs, and support health education initiatives promoting safe practices and voluntary blood donation from low-risk individuals.

Abbreviations

HBV	Hepatitis B virus
HCV	Hepatitis C Virus
HIV	Human Immunodeficiency Virus
BTS	Blood Transfusion Services
TTIs	Transfusion-Transmitted Infections
AOR	Adjusted Odd Ratio
ELISA	Enzyme-Linked Immunosorbent Assay
RPR	Rapid plasma regain
BBL	Blood Bank Laboratory
CI	Confidence Interval

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Author contributions

This study was conducted in conjunction with all authors. MD: planned and designed the study, as well as contributed to data administration and paper writing. DMM: design, data collecting, analysis, and interpretation, in manuscript edition. MG: Design, data collection, interpretation, and preliminary article write-up and Analysis. TY: Design, Data Collection, reviewed the draft paper, and contributed to analysis of the results. All writers contributed adequately to the work and accepted responsibility for the relevant sections of the text.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Ethical clearance for this study was obtained from the Institutional Review Board (IRB) of the GT Technology collage (Reference No. GTC/21/08/2022). Informed consent was not used because the study was retrospective. But by keeping the information that was extracted private and making sure that it could only be utilized for study-related objectives, the data's secrecy was preserved. The participant's informed consent waiver was authorized by the institutional ethical review board of the GT Technology College in Gondar, Ethiopia. The study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki (2013 version).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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